

MANCHIKANTI SURYA VARDHAN

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Professional Summary

M.Tech VLSI Design student with hands-on experience in RTL-to-GDSII ASIC implementation and custom IP core development. Proficient in Verilog HDL, Cadence EDA toolchain (Genus, Innovus, NCLaunch), and Xilinx Vivado, Quartz Prime for SoC and FPGA development. Experienced in synthesis, place-and-route, STA, physical design optimization, and functional verification methodologies. Strong understanding of Digital Design, CMOS VLSI Design, RISC-V architecture. Seeking opportunities in ASIC Design, Physical Design, CAD/EDA, and semiconductor technology development.

Technical Skills

HDL / Languages	Verilog HDL, SystemVerilog (Basics), TCL , Linux Shell Scripting
EDA Tools	Cadence Genus, Cadence Innovus, Cadence NCLaunch, QuestaSim, ModelSim, Vivado, Vitis AI, Vitis HLS
ASIC Flow	RTL-to-GDSII Flow, Logic Synthesis, Floorplanning, Placement, CTS, Routing, STA, Timing Closure, DRC/LVS Basics
FPGA / SoC Architecture	Xilinx ZCU104 MPSoC, IP Integrator, DPU Integration, Altera DE10 RISC-V (RV32I), MAC Unit Design, FIR Filter Design, SoC/IP Integration
Core Concepts	Digital Logic Design, CMOS VLSI Design, Digital IC Design, ASIC Design Flow, Physical Design

Projects

32-bit RISC-V Processor with Integrated MAC Unit — RTL-to-GDSII ASIC Flow

- Designed and implemented a 32-bit single-cycle RISC-V processor with a Multiply-Accumulate (MAC) Unit in Verilog HDL.
- Executed the complete **RTL-to-GDSII ASIC design flow**, including RTL coding, functional simulation, synthesis, floorplanning, placement, clock tree synthesis (CTS), routing, and timing closure.
- Performed logic synthesis using **Cadence Genus** and physical implementation using **Cadence Innovus** with gpd90 (90nm) Process Design Kit (PDK) technology library.
- Applied timing constraints using SDC and performed **Static Timing Analysis (STA)** to achieve positive timing slack and stable timing closure.

FPGA-Based Drone Detection and Classification System using YOLOv8n

- Implemented a hardware-accelerated drone detection pipeline on FPGA using lightweight **YOLOv8n** architecture for real-time edge AI inference.
- Trained and fine-tuned the YOLOv8n model on a custom drone dataset, improving detection precision by approximately **5.3%**.
- Exported trained AI model to **ONNX** format and deployed it using Xilinx Vitis AI DPU framework on ZCU104 MPSoC.
- Designed a custom FPGA **IP Core** for post-processing and **Non-Maximum Suppression (NMS)** to filter overlapping detections and improve accuracy. Integrated custom IP and validated the complete design on hardware.

VLSI Design of 2D FIR Filter Architecture using CSD

- Designed a 2D FIR Filter Architecture using **Canonical Signed Digit (CSD)** representation to eliminate conventional multipliers.
- Implemented multiplier-less DSP architecture using optimized shift-and-add operations, reducing hardware complexity and power consumption.
- Generated and optimized FIR filter coefficients using MATLAB and implemented the architecture using Verilog HDL in Xilinx Vivado.
- Achieved improved area and power efficiency compared to traditional multiplier-based FIR filter implementations.

Education

Vellore Institute of Technology, Chennai Master of Technology (M.Tech) in VLSI Design CGPA: 8.31	2025 – Present
CVR College of Engineering, Hyderabad Bachelor of Technology (B.Tech) in Electronics and Communication Engineering CGPA: 8.23	2019 – 2023
Narayana Junior College, Hyderabad Intermediate (MPC) Percentage: 89.9%	2017 – 2019
Narayana E-Techno School, Hyderabad SSC (10th Grade) CGPA: 9.3	2017

Certifications

- **The Complete Guide to SystemVerilog Assertions (SVA)** — Srinivasan Venkataramanan Ongoing
- **ASIC Flow, Logic Synthesis, STA, Physical Design** — Cadence Genus, Tempus, Innovus & Synopsys
ICC2 (Udemy - VLSI Mentor) Ongoing
- **Certified in Verilog HDL — Hands-On** — Maven Silicon 2025
- **Certified in CMOS Digital VLSI Design** — NPTEL / SWAYAM 2025